

Brief Review

1. Consider a particle moving with velocity $v(t) = \sqrt{t} + 3$ ft/sec from $t = 0$ to $t = 3$ seconds.
 - (a) Make a very rough guess about the distance traveled by the particle.
 - (b) Use a Riemann sum with 4 subdivisions and left endpoints to estimate this distance.
 - (c) Repeat using 6 subdivisions. What do you notice?
 - (d) Use a definite integral to find the exact distance traveled by the particle. Compare with your estimates.
2. Consider the area under the graph of $y = \sqrt{x} + 3$ between $x = 0$ and $x = 2$.
 - (a) Use a Riemann sum with 4 subdivisions and left endpoints to estimate this area. Draw the region and the 4 rectangles.
 - (b) Use a definite integral to find the exact area.
 - (c) Explain the connection between Riemann sums, definite integrals, distance, and area.
3. Find $\frac{dy}{dx}$ if
 - (a) $y = x^7 - 5x^{\frac{3}{2}} + \frac{4}{x} - 3$
 - (b) $y = \sqrt{\cos \sqrt{x}}$
 - (c) $y = e^{4x} \tan x^2$
 - (d) $y = \frac{3x^2 \csc x}{\sin^{-1} x}$
4. Evaluate the following integrals.
 - (a) $\int \sqrt{x} + x^{-\frac{3}{2}} dx$
 - (b) $\int \frac{\sin x dx}{(1 + 2 \cos x)^2}$
 - (c) $\int \sec^2(x) \sec(\tan x) dx$